Evaluating Very Long-Term Conversational Memory of LLM Agents

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Abstract

Existing works on long-term open-domain dialogues focus on evaluating model responses within contexts spanning no more than five chat sessions. Despite advancements in long-context large language models (LLMs) and retrieval augmented generation (RAG) techniques, their efficacy in *very* long-term dialogues remains unexplored. To address this research gap, we introduce a machine-human pipeline to generate high-quality, *very* long-term dialogues by leveraging LLM-based agent architectures and grounding their dialogues on personas and temporal event graphs. Moreover, we equip each agent with the capability of sharing and reacting to images. The generated conversations are verified and edited by human annotators for long-range consistency and grounding to the event graphs. Using this pipeline, we collect LoCoMo, a dataset of *very* long-term conversations, each encompassing 300 turns and 9K tokens on avg., over up to 35 sessions. Based on LoCoMo, we present a comprehensive evaluation benchmark to measure long-term memory in models, encompassing question answering, event summarization, and multi-modal dialogue generation tasks. Our experimental results indicate that LLMs exhibit challenges in understanding lengthy conversations and comprehending long-range temporal and causal dynamics within dialogues. Employing strategies like long-context LLMs or RAG can offer improvements but these models still substantially lag behind human performance.[1]

1 Introduction

Despite recent advancements in dialogue models based on LLMs for extended contexts (Bertsch et al., 2024; Xiao et al., 2023), as well as the integration of retrieval augmented generation (RAG) techniques (Shuster et al., 2021; Ram et al., 2023; Shi et al., 2023), there is still a need for thorough evaluation of their efficacy in handling very long conversations. Indeed, studies in long-term open-domain dialogues have concentrated on assessing model responses within limited contexts e.g., ∼1K tokens over five chat sessions (Xu et al., 2022; Jang et al., 2023; Zhang et al., 2023). This long term evaluation is crucial for refining engaging chatbots capable of remembering key information from past interactions, to generate empathetic, consistent, and useful responses.

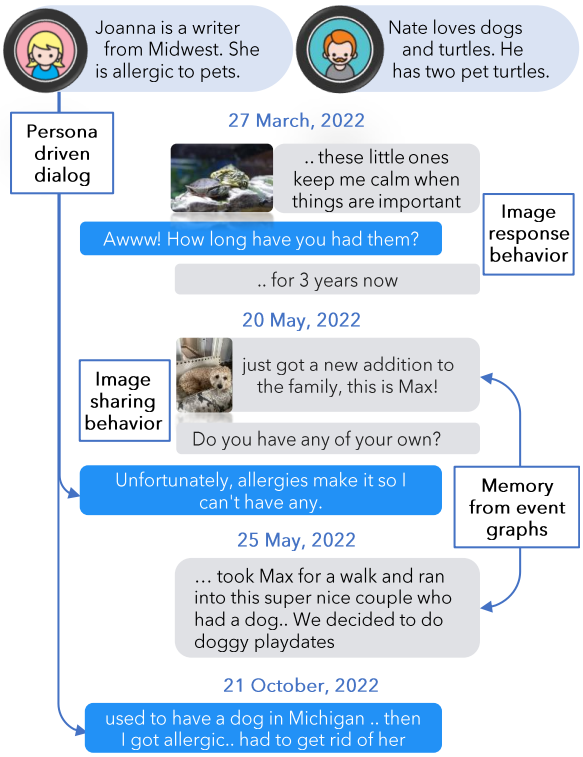

Figure 1: **An example in LoCoMo.** Dialogs are steered by the speakers' personas and corresponding events e.g., Joanna's responses are consistent with her pet allergies. For Nate, the event *got a new dog* is followed by a *playdate with neighbor's dog*, showcasing long-term memory. Multimodal dialog is enabled with image-sharing and image-response behaviors.

[1]Code and data to be available at https://snap-research.github.io/locomo

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Dataset	Avg. turns per conv.	Avg. sessions per conv.	Avg. tokens per conv.	Time Interval	Multimodal	Collection
MPChat (Ahn et al., 2023)	2.8	1	53.3	-	✓	Reddit
MMDialog (Feng et al., 2023)	4.6	1	72.5	-	✓	Social media
Daily Dialog (Li et al., 2017)	7.9	1	114.7	-	✗	Crowdsourcing
SODA (Kim et al., 2023)	7.6	1	122.4	-	✗	LLM-generated
MSC (Xu et al., 2022) (train; 1-4 sessions)	53.3	4	1,225.9	few days	✗	Crowdsourcing
Conversation Chronicles (Jang et al., 2023)	58.5	5	1,054.7	few hours - years	✗	LLM-generated
LoCoMo (ours)	304.9	19.3	9,209.2	few months	✓	LLM-gen. + crowdsourc.

Table 1: **Statistics of LoCoMo** compared to existing dialog datasets. The average length of a conversation in LoCoMo is 9x that of MSC (Xu et al., 2022), distributed over 6x more turns and 4x more sessions (on average).

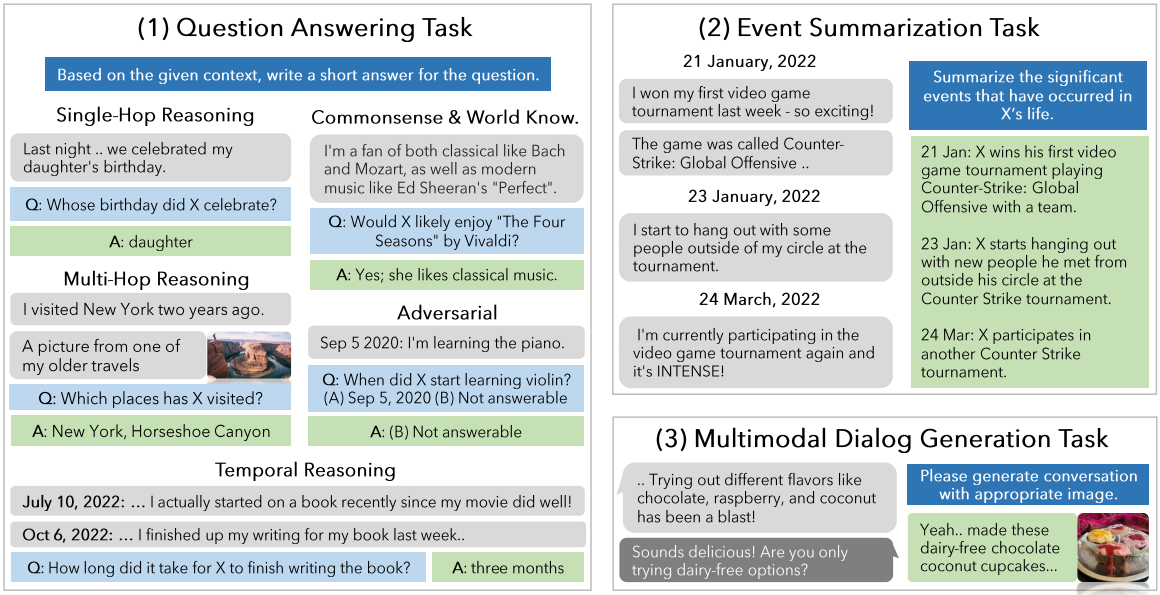

Figure 2: **Overview of our evaluation framework**. We propose three tasks: question answering, event summarization and multimodal dialog generation to evaluate models' comprehension in very long-term dialogues.

To this end, we present the first study of very long-term open-domain multi-modal dialogues, closely mirroring real-world online interactions, collected via a human-machine pipeline where we first use LLM-based generative agents to generate conversations and then ask human annotators to fix any long-term inconsistencies in the conversations. Specifically, drawing on the understanding that real-world conversations are a complex blend of collective memories (Assmann and Czaplicka, 1995; Hirst and Manier, 2008), individual viewpoints (Hirst et al., 2018), external influences (Hirst and Echterhoff, 2012), and the unique persona of the speakers (Pruitt and Grudin, 2003; Cooper, 1999; Zhou et al., 2020; Shum et al., 2020), we create *very long-term* dialogues based on LLM agent with the following features: (1) a unique persona (§3.1); (2) a timeline of causally interlinked events in their lives (§3.2); and (3) *reflect & response* mechanism to respond based on dialogue history (like in Park et al. (2023)) and *image sharing & image reaction* behavior which sends or reacts to images (§3.3). Finally, human annotators fix long-range inconsistencies in dialogues, remove irrelevant images, and verify the grounding of dialogs to events (§3.4). With this pipeline, we create LoCoMo, a dataset of 50 *very long-term* dialogues, each consisting of 300 turns and 9K tokens on avg., over up to 35 sessions (see Figure 1 and Table 1).

Conventional approaches for evaluating conversational agents in open-domain dialogues involves directly evaluating the agent response based on past dialogue history. It often employs lexical overlap (Papineni et al., 2002) and semantic overlap (Zhang et al., 2019) between ground truth and the agent response, or consistency (Ghazarian et al., 2022), contradiction (Nie et al., 2021; Welleck et al., 2019), and empathy (Zhang et al., 2021a, 2022) of the agent response. However, these evaluation metrics are not well-suited for directly assessing the agent's comprehension of long-term contexts.

In this study, we present a holistic evaluation framework to assess an agent's proficiency in man-

aging and responding within long-term contexts (see Figure 2). First, agents need to "recall" past context correctly to integrate relevant information into future responses. We present a direct examination of their *memory* via a *question answering* (QA) task (§4.1). We classify questions into five distinct reasoning types to evaluate memory from multiple perspectives: single-hop, multi-hop, temporal, commonsense or world knowledge, and adversarial. Second, agents also need to recognize long-range causal and temporal connections in the dialogues to generate empathetic and relevant responses. We propose a measurement of their causal and temporal understanding with an *event graph summarization* task (§4.2). In this task, the event graphs linked to each LLM speaker serve as the correct answers, and models are tasked with extracting this information from the conversation history. Third, conversational agents need to utilize relevant context recalled from past conversations to generate responses that are consistent with the ongoing narrative. We assess this ability via the *multi-modal dialog generation* task (§4.3).

We present extensive experimental results on the LoCoMo benchmark using instruction-based LLMs, long-context LLMs, and RAG techniques (§5). Our findings include:

(1) Long-context LLMs and RAG demonstrate effectiveness in QA tasks, improving 'memory' capabilities of LLMs (with improvements ranging from 22-66%), but still significantly lag behind human levels (by 56%), especially in temporal reasoning, (by 73%);

(2) long-context LLMs demonstrate significant difficulty with adversarial questions in the QA task, showing a performance that is 83% lower than the base model. They are especially prone to misassigning dialogs or events to the wrong speaker. Moreover, they show poor performance on *event graph summarization*, lagging behind the base model by 14%, indicating that they may grasp the factual elements within the entire conversation but do not accurately comprehend the context; and

(3) RAG offers a balanced compromise, combining the accuracy of short-context LLMs with the extensive comprehension of wide-context LLMs, and does particularly well when dialogues are transformed into a database of assertions (*observations*) about each speaker's life and persona.

2 Related Work

Long-term Dialogue. Recent approaches involve retrieving historical context from a range of previous dialogues and reasoning over retrieved segments in a temporal order (Lee et al., 2023b; Lu et al., 2023; Zhong et al., 2023; Liang et al., 2023) and/or using events to scaffold the dialogues (Jang et al., 2023; Zhang et al., 2023) to enable consistency in long-term conversations. Some limitations of such frameworks are: (1) The accuracy of retrieval can be compromised, as the retrieval model is generally trained on tasks focusing on semantic similarity rather than specifically on such dialogues. Additionally, real-world dialogues often feature co-references and missing content (*i.e.,* anaphora) (Anantha et al., 2021), which further complicate the retrieval process (Mallen et al., 2023; Gao et al., 2023b; Liu et al., 2023); (2) Challenges arise in reasoning over retrieved documents, especially when the model struggles to identify the correct context among the retrieved data (Liu et al., 2024); (3) Reasoning over time intervals presents challenges. For example, the way a system responds about past events can vary depending on the amount of time that has passed since the last conversation (Zhang et al., 2023; Jang et al., 2023). Therefore, it is essential to have conversations of considerable length, as well as a systematic evaluation framework, to accurately assess the effectiveness of approaches to long-term dialogue generation. We design a long-term conversation generation pipeline based on retrieval augmentation and events graphs and propose a framework for evaluating long-term dialog agents.

Multi-modal Dialogue. Multi-modal dialogue primarily consists of two types of tasks: image-grounded dialogue and image-sharing dialogue. The image-grounded dialogue task is centered around responding to questions (Antol et al., 2015; Das et al., 2017; Kottur et al., 2019) or creating natural conversations related to specific images (Mostafazadeh et al., 2017; Shuster et al., 2020; Meng et al., 2020; Zheng et al., 2022). Conversely, the image-sharing dialogue task focuses on selecting images that semantically align with the provided dialogue context (Zang et al., 2021; Feng et al., 2023; Lee et al., 2023c). We use a method from the image-sharing dialogue task to create multimodal dialogs which are then evaluated as an image-grounded dialogue task.